

PRACTICAL - 10

Title:

- To understand the installation and configuration of operating systems.
- To study different variants of operating systems such as Windows and Linux.

Requirements:

- Computer system (PC/Laptop)
- Bootable USB/DVD with OS (Windows/Linux)
- Minimum 4GB RAM (recommended 8GB)
- At least 20GB free disk space

Theory:

An Operating System (OS) is an essential system software that acts as an interface between the user and the computer hardware. It controls and coordinates the use of hardware resources such as CPU, memory, storage devices, and input/output devices. The OS provides a platform for running application programs and ensures smooth execution of tasks.

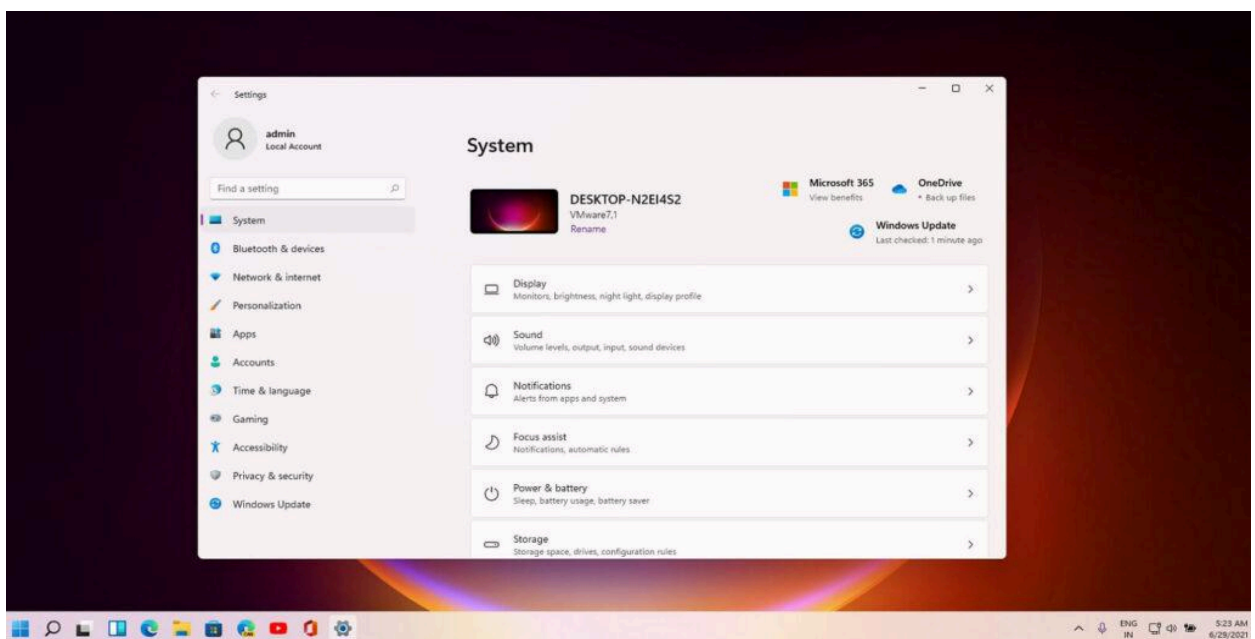
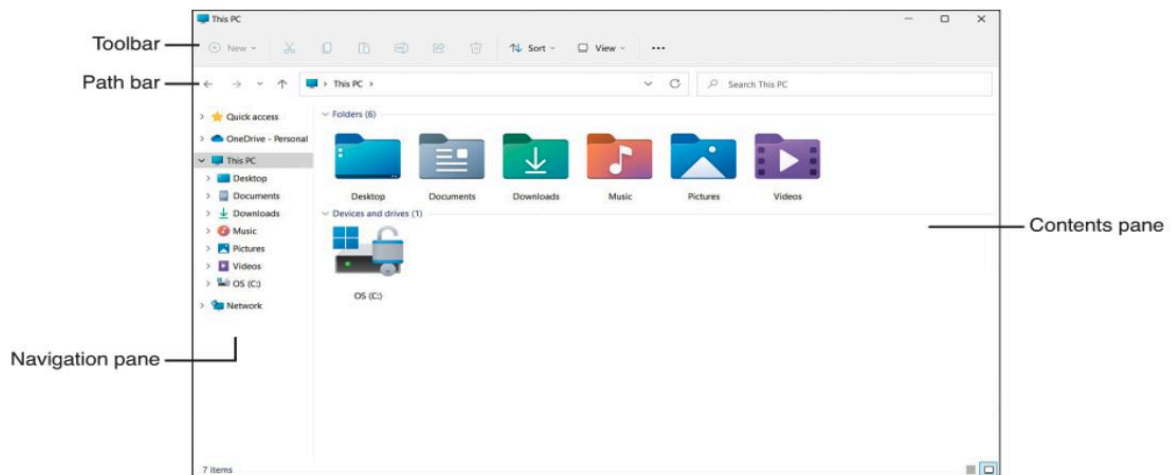
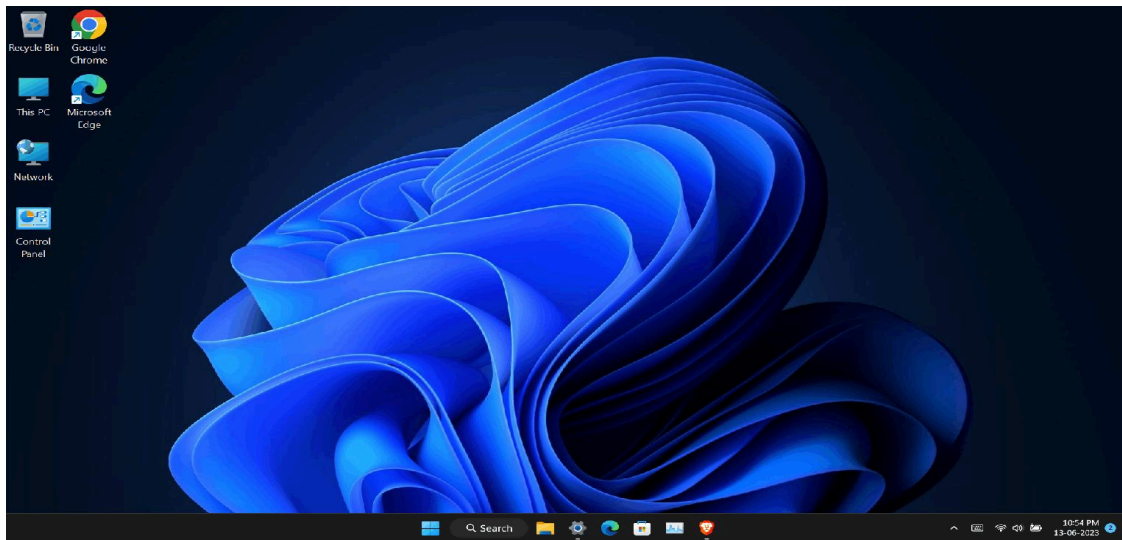
The primary functions of an operating system include:

- Process Management: Handles creation, scheduling, and termination of processes.
- Memory Management: Allocates and deallocates memory to programs efficiently.
- File System Management: Organizes and manages files on storage devices.
- Device Management: Controls hardware devices using drivers.
- Security and Access Control: Protects data and system resources.
- User Interface: Provides interaction through GUI (Graphical User Interface) or CLI (Command Line Interface).

Operating systems can be classified into different types such as:

- Batch OS
- Time-sharing OS
- Distributed OS
- Real-time OS
- Multi-user and multitasking OS

WINDOWS OPERATING SYSTEM:



The Windows OS, developed by Microsoft, is one of the most widely used operating systems in the world. It is known for its user-friendly interface and extensive software compatibility.

Features:

- Graphical User Interface (GUI) with icons, menus, and taskbar
- Supports multitasking and multi-user environment
- Plug and Play hardware support
- Built-in security features like Windows Defender
- Wide compatibility with software and hardware devices

Advantages:

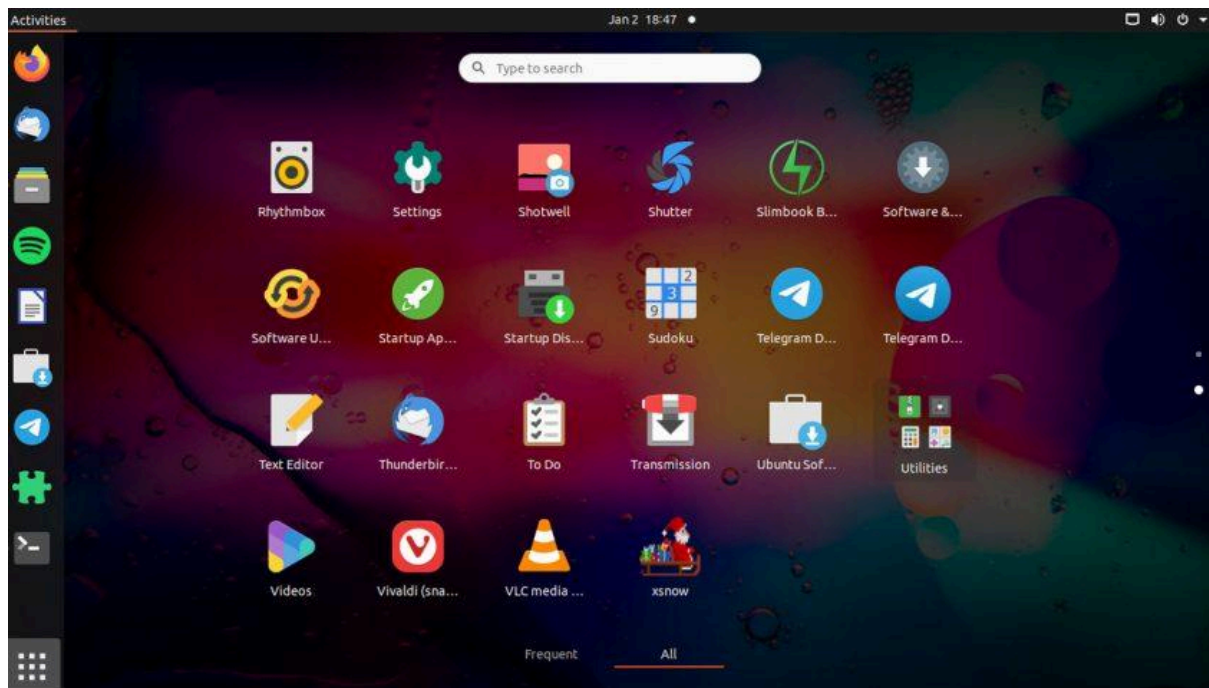
- Easy to use and learn
- Large community and support
- Compatible with most applications and games

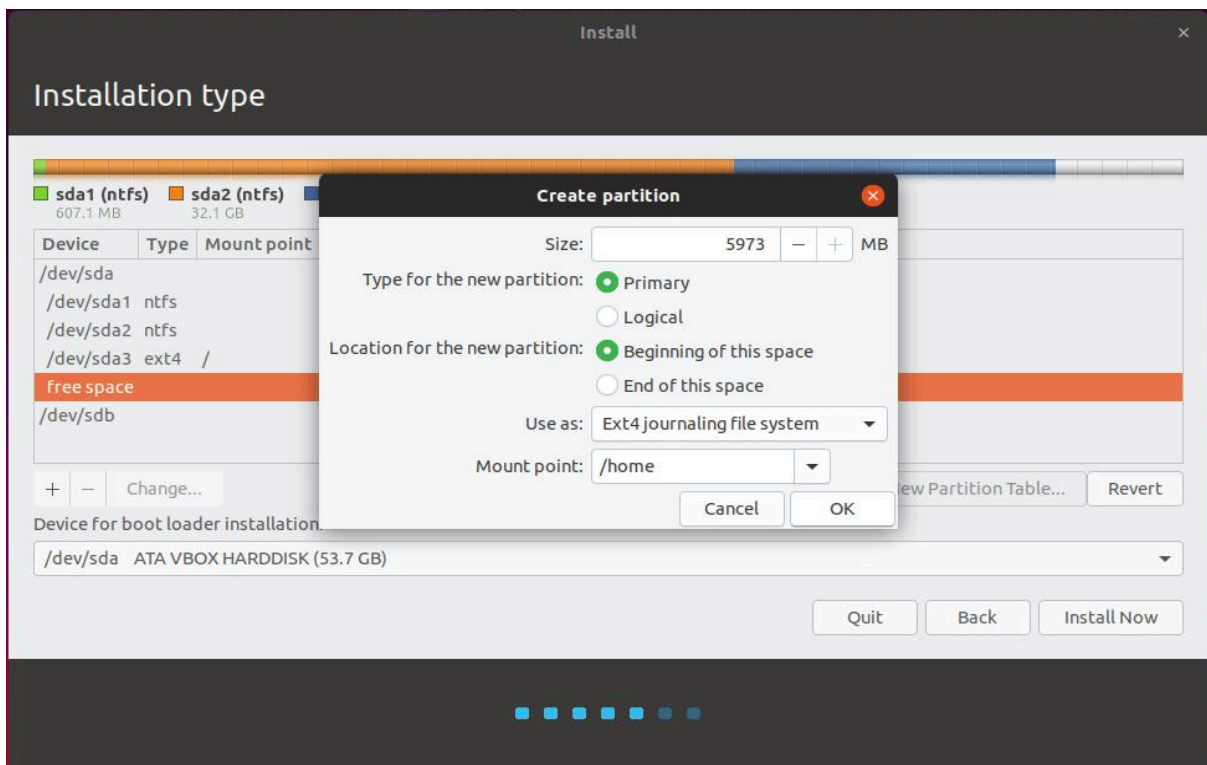
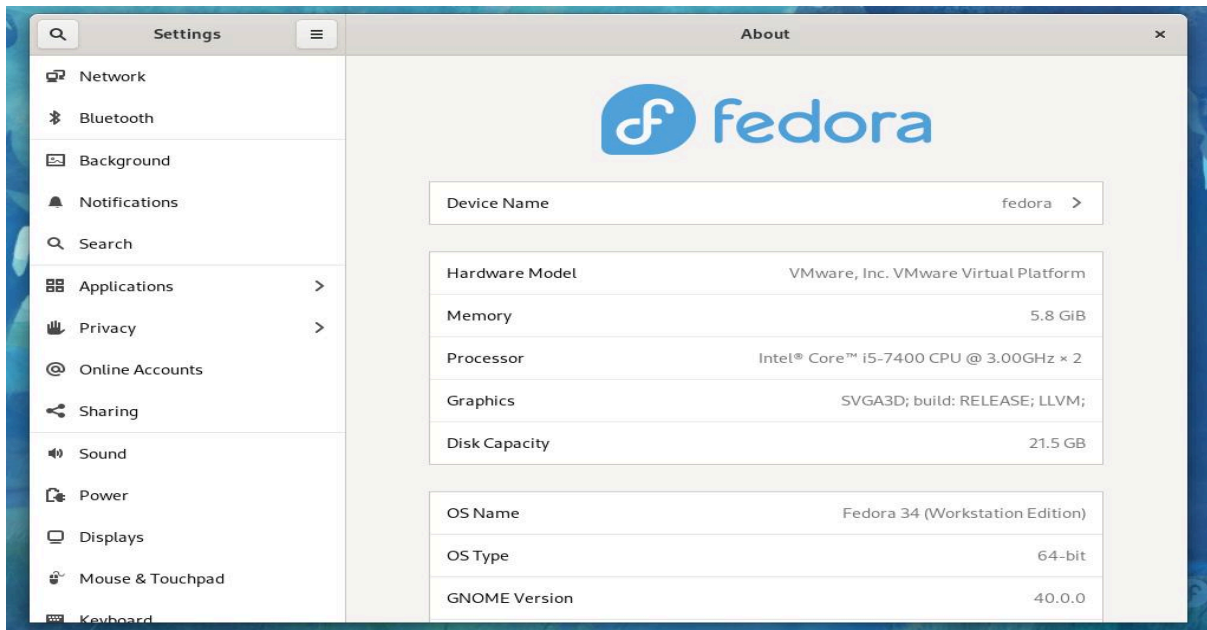
Disadvantages:

- Paid license required
- More vulnerable to malware compared to Linux

Examples: Windows 10, Windows 11

Linux Operating System:





The Linux OS is an open-source operating system based on the Unix architecture. It is widely used in servers, embedded systems, and increasingly in personal computers.

Popular distributions include Ubuntu and Fedora.

Features:

- Open-source and free to use
- Strong security and stability
- Multi-user and multitasking capability
- Command Line Interface (CLI) along with GUI
- Highly customizable

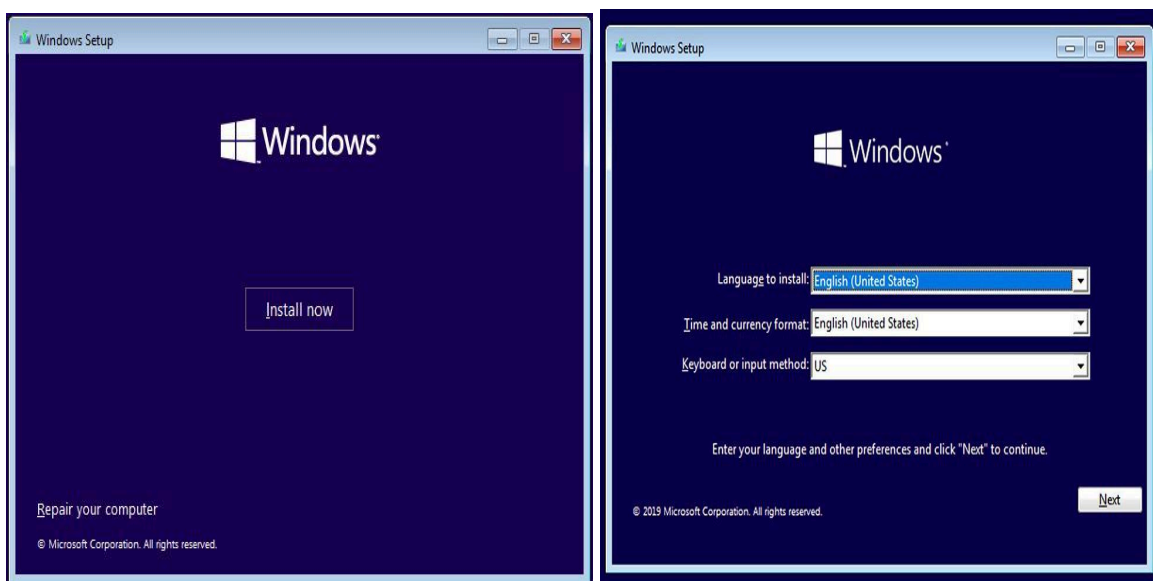
Advantages:

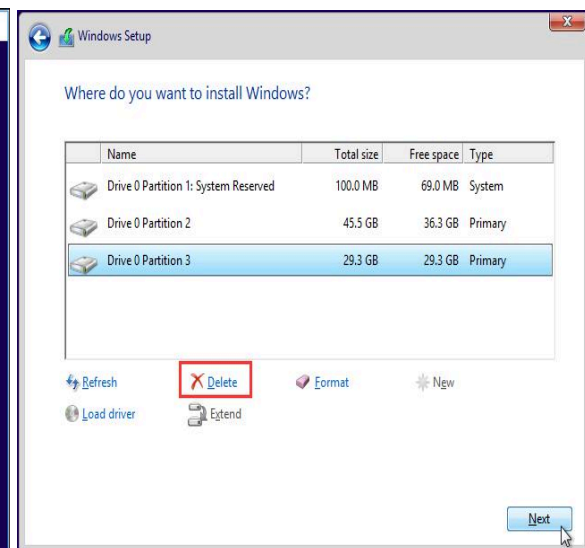
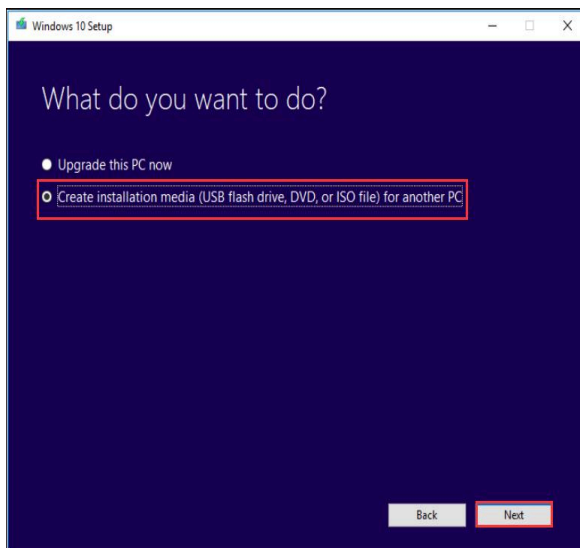
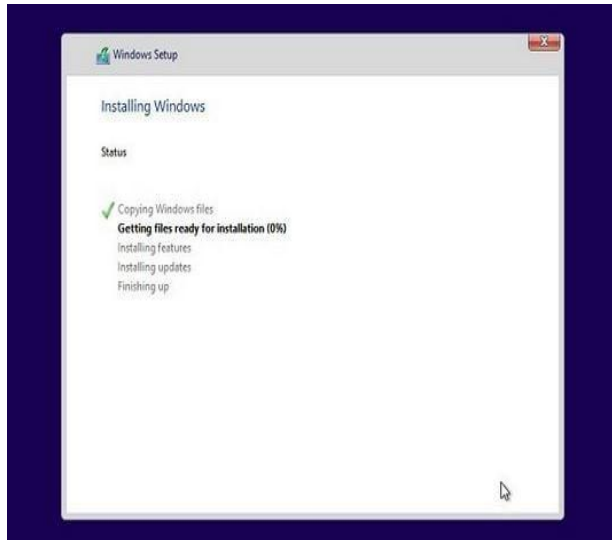
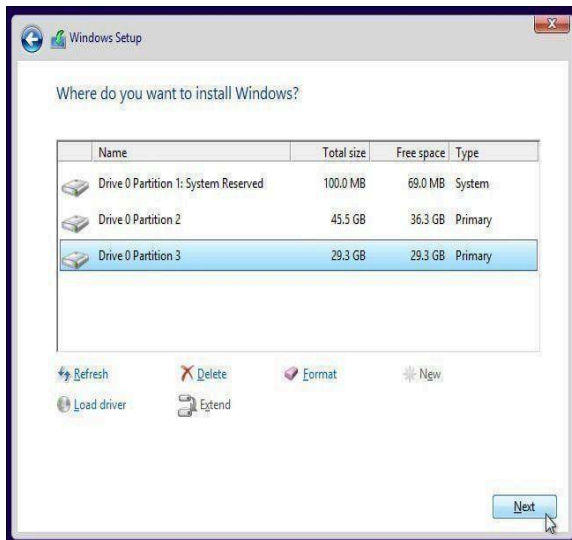
- Free and open-source
- Highly secure and less prone to viruses
- Efficient performance even on low-end systems

Disadvantages:

- Steeper learning curve for beginners
- Limited support for some proprietary software

Installation of Windows OS:

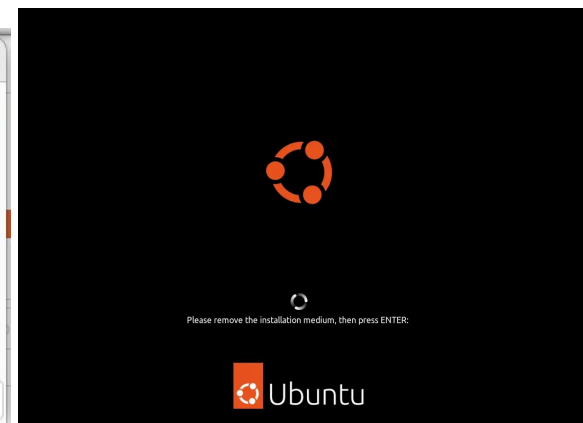
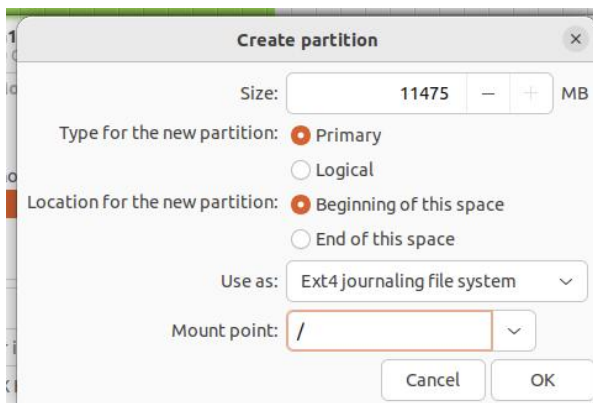
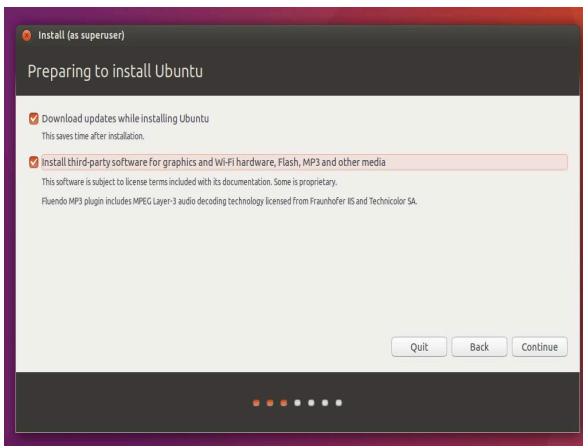
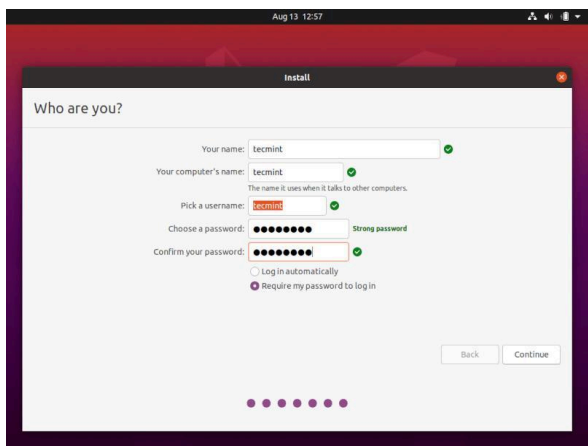
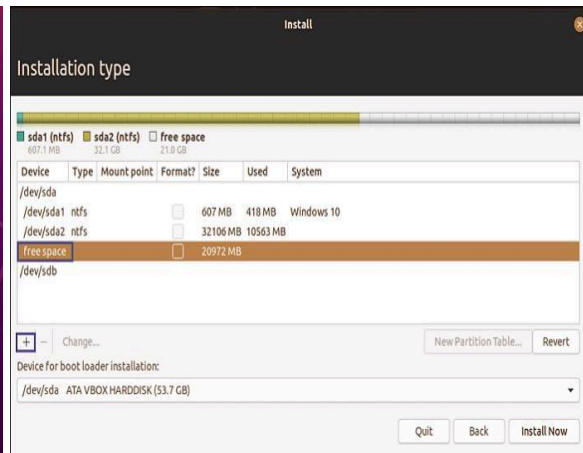
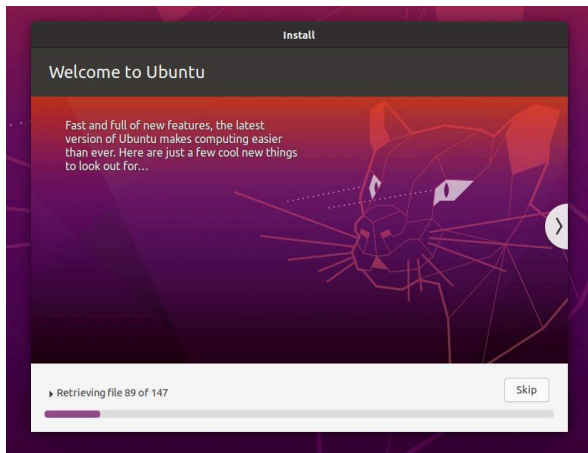




Steps:

1. Insert bootable USB/DVD containing Windows OS.
2. Restart the system and enter BIOS/Boot Menu (using F2/F12/DEL key).
3. Select boot device (USB/DVD).
4. Windows Setup screen appears.
5. Choose language, time, and keyboard preferences.
6. Click on **Install Now**.
7. Enter product key (or skip).
8. Select installation type (**Custom Installation**).
9. Create/format disk partitions.
10. Select partition and click **Next**.
11. Installation process begins.
12. System restarts automatically.
13. Set username, password, and preferences.

Installation of Linux OS (Ubuntu/Fedora):



Steps:

1. Insert bootable USB with Linux OS.
2. Restart system and boot from USB.
3. Select Try or Install Linux option.
4. Choose language and keyboard layout.
5. Select Install OS.
6. Choose installation type:
7. Erase disk OR

8. Install alongside Windows (Dual Boot)
9. Partition disk if required.
10. Set timezone.
11. Create username and password.
12. Wait for installation to complete.
13. Restart system and remove USB.

Disk Partitioning:

- Disk partitioning divides storage into sections.
- Common partitions:
 - Primary Partition (OS installation)
 - Extended Partition
 - Logical Drives
- In Linux:
 - Root /
 - Swap area
 - Home /home

Basic Configuration:

After installation:

- Create user account and password
- Set date and time
- Install system updates
- Install necessary drivers/software
- Configure system settings

Result:

- Successfully studied different types of operating systems.
- Successfully installed and configured Windows and Linux operating systems.